Gradient Descent for Machine Learning — Lecture 4: Optimizat

Source: synthetic_gd · **Duration:** 2:17 · **Faithfulness:** 83% of key points supported

Summary

Welcome back. Today we study gradient descent, the workforce optimization algorithm behind most of machine learning. Choosing the learning rate matters a lot. If it is too large, the iterates oscillate or even diverge. Stochastic gradient descent estimates it from a small mini batch of examples. The estimate is noisy, but each step is much cheaper, so overall training is faster.

Key points

- [!] Welcome back. 0:00 _[both]_
  - [!] *Not supported by the transcript/slides — check before relying on it.*
- [ok] Today we study gradient descent, the workforce optimization algorithm behind most of machine learning. 0:00 _[both]_
- [ok] Choosing the learning rate matters a lot. 0:46 _[both]_
- [ok] If it is too large, the iterates oscillate or even diverge. 0:46 _[both]_
- [ok] Stochastic gradient descent estimates it from a small mini batch of examples. 1:08 _[both]_
- [ok] The estimate is noisy, but each step is much cheaper, so overall training is faster. 1:08 _[both]_

Detailed summary

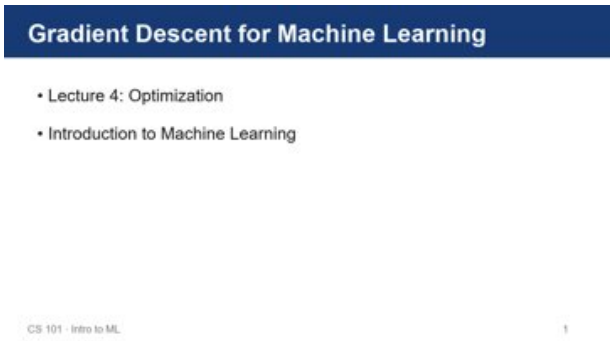
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Chapters

- 0:00 Gradient Descent for Machine Learning — Lecture 4: Optimizat
- 0:46 The Update Rule — W <- W eta grad L(w) | eta = learning rate
- 1:08 Choosing the Learning Rate — Too large: divergence oscillati

Study notes

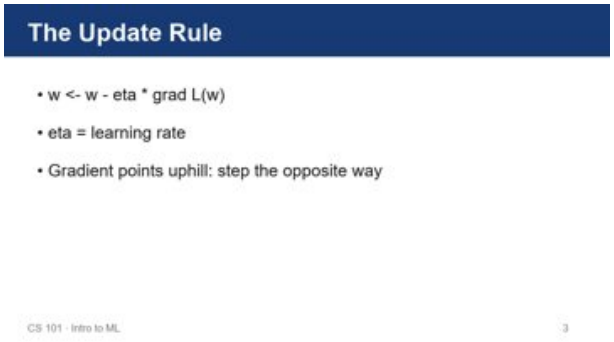
Gradient Descent for Machine Learning — Lecture 4: Optimizat



- Welcome back.
- Today we study gradient descent, the workforce optimization algorithm behind most of machine learning.

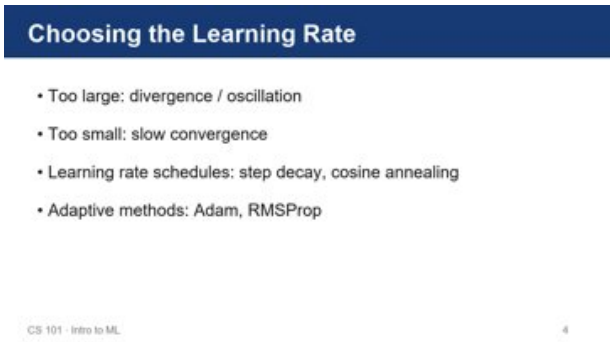
• We will see what it does, why it works, and how to choose its settings.

The Update Rule — W <- W eta grad L(w) | eta = learning rate



• Choosing the learning rate matters a lot.
• If it is too large, the iterates oscillate or even diverge.
• If it is too small, training converges very slowly.

Choosing the Learning Rate — Too large: divergence oscillati



• Stochastic gradient descent estimates it from a small mini batch of examples.
• The estimate is noisy, but each step is much cheaper, so overall training is faster.
• To summarize, gradient descent repeatedly steps along the negative gradient, the learning rate controls the step size, and stochastic gradient descent makes each step cheap by using mini batches.

Glossary

• **Gradient Descent for Machine Learning —** — Welcome back.
• **The Update Rule — W <- W eta grad L(w) |** — Choosing the learning rate matters a lot.
• **Choosing the Learning Rate — Too large** — Stochastic gradient descent estimates it from a small mini batch of examples.

Revision questions

Q1. Which statement was made in the lecture? (1)
- A) Welcome back.
- B) None of these
- C) All of these
- D) Not discussed

Answer: A — Stated in the lecture.

Q2. Which statement was made in the lecture? (2)
- A) Today we study gradient descent, the workforce optimization algorithm behind most of machine learning.

- B) None of these
- C) All of these
- D) Not discussed

Answer: A — Stated in the lecture.

Q3. Which statement was made in the lecture? (3)

- A) We will see what it does, why it works, and how to choose its settings.
- B) None of these
- C) All of these
- D) Not discussed

Answer: A — Stated in the lecture.

Q4. Which statement was made in the lecture? (4)

- A) Choosing the learning rate matters a lot.
- B) None of these
- C) All of these
- D) Not discussed

Answer: A — Stated in the lecture.

Q5. Which statement was made in the lecture? (5)

- A) If it is too large, the iterates oscillate or even diverge.
- B) None of these
- C) All of these
- D) Not discussed

Answer: A — Stated in the lecture.

S1. Explain: Welcome back.
Model answer: Welcome back.

S2. Explain: Today we study gradient descent, the workforce optimization algorithm behind mos
Model answer: Today we study gradient descent, the workforce optimization algorithm behind most of machine learning.

S3. Explain: We will see what it does, why it works, and how to choose its settings.
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S4. Explain: Choosing the learning rate matters a lot.
Model answer: Choosing the learning rate matters a lot.

S5. Explain: If it is too large, the iterates oscillate or even diverge.
Model answer: If it is too large, the iterates oscillate or even diverge.

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